

ALIKHAN KABDUALY

Building AI products at 16. From Kostanay, Kazakhstan.

hi@alikhandev.com alikhandev.com github.com/Krazher220-Ceo Kostanay, Kazakhstan

ABOUT

I'm Alikhan, a 16-year-old developer from Kostanay, Kazakhstan. I build AI-powered products — computer vision models, progressive web apps and Telegram bots — taking them from concept to deployment. I've competed in hackathons and pitch sessions across the Astana Hub ecosystem in Kostanay, Petropavlovsk and Almaty. My current project is Smart Green Qostanai, a service for monitoring urban tree health.

PROJECTS AND VENUES

Jasyl2026-08 · 2 days

AI monitoring of Kostanay's urban trees

Venue: Qostanai Smart City Hackathon · case No. 1

Role: Development and ML — the app and its cross-platform builds, model training, the inference service, the database and its access rules, offline sync

Stack: Vite · React 18 · TypeScript · PyTorch · MobileNetV3-Large · ONNX Runtime Web · Supabase · Leaflet · Capacitor 6 · Electron 33

- 343 photos in the dataset, split by tree 268 / 75
- 54,7% best validation accuracy
- 0,55 macro F1 across six classes
- 16,8 MB ONNX model for CPU inference

QA Vision2025-11 · 3 days

Paint-quality control on car bodies, from a photograph

Venue: Qostanai AI-Sana Industry Hackathon · Allur Challenge

Role: Development and architecture — the dashboard, the login service and API, the pluggable AI module system, the detection registry and KPI maths, the exports, roles and access control

Stack: Python · Streamlit · Plotly · FastAPI · Roboflow · Ultralytics YOLO · PyTorch · OpenCV · pandas · PyJWT · Power BI · Looker Studio

- 7 787 lines of Python in the project
- 4 kinds of AI module behind one interface
- 0,70 default confidence threshold
- 3 access roles: superadmin, plant manager, worker

KZ UniVerse2025-12 · 24 hours

A single platform for Kazakhstan's universities

Venue: IT Fest 2025 · Hackathon track

Role: Development — the catalogue and university pages, comparison, analytics, the AI provider chain, profile and portfolio, the Telegram bot, and the demo rig for the pitch

Stack: Next.js 14 · TypeScript · Tailwind CSS · Framer Motion · Recharts · Ollama · Google Gemini 2.5 Flash Lite · Telegram Bot API

- 15 universities in the base
- 19 study programmes
- 3 assistant modes with fallbacks
- 24 ч for the whole track

A marketplace of ready-made industrial IoT modules

Venue: Startup Battle, IT Fest 2025 · top 30

Role: Product and development — the marketplace front, the dashboards, the Eco module, login and protected code preview, the project economics, the pitch and the answers to the jury

Stack: JavaScript · HTML/CSS · Google OAuth · IoT · Power BI

- 30 teams in the in-person Startup Battle final
- 3 cities where the project was shown
- 2024 the year of the first venue
- 4 revenue streams in the model

TIMELINE · 2024 → 2026

Idea battles and pitch sessions at Qostanai Hub

2024 · Kostanay

First outings with NPAI: learned to fit an idea into a short pitch, and to hear questions about money rather than code.

Pitch venue

2024–2025 · Petropavlovsk

A conversation with people from outside my own city, where the question of who actually buys this lands hardest.

Qostanai AI-Sana Industry Hackathon · Allur Challenge

2025 · Kostanay

1–3 November at Dulatov University. The first product built against a real manufacturing case — QA Vision.

IT Fest 2025 — two tracks

2025 · Almaty

5–6 December. A 24-hour hackathon with KZ UniVerse and Startup Battle with NPAI: top-30 teams, pitched in person.

Qostanai Smart City Hackathon · case No. 1

2026 · Kostanay

13–14 August. Jasy!l: our own dataset, a fine-tuned model and a working offline app in two days.

STACK

CONFIDENT	Python · JavaScript · HTML/CSS · FastAPI · aiogram · Git
USED IN PROJECTS	PyTorch · TensorFlow · MobileNetV2 · EfficientNet · YOLOv8 · Supabase · Leaflet.js · PWA · Render · Docker
LEARNING	React · n8n · Ollama · LM Studio

CERTIFICATES

- ШАГ · School of Active Citizens 2025–05
Personal; the project was built with city budget funding
- Pizza Pitch · Qostanai Hub 2025–08
Third place in the startup idea contest
- Qostanai AI-Sana Industry Hackathon: Allur Challenge 2025–11
Personal; signed by Dulatov University, the ITU, Allur and Qostanai Hub
- IT Fest 2025 2025–12
Personal; national festival, Almaty
- Qostanai Smart City Hackathon 2026–08
Issued to the team, not to a person

EDUCATION

- School-Lyceum No. 1, Kostanay — maths stream
- Music school — completed

LANGUAGES

Kazakh · Russian · English